
University of Waterloo
School of Computer Science
CS 486/686, **SAMPLE** Final Examination
Winter 2025

- Instructor: Jesse Hoey and Victor Zhong
- Date: Sometime Before April 22nd, 9:00am
- Location: Wherever
- Time: Whenever
- There are 5 questions on this exam
- There are 6 pages in this exam
- There are a total of 100 marks on the exam
- You have 150 minutes to complete the exam
- Non-programmable calculators are allowed
- Write your answers on a separate sheet. On the **REAL** final, you will write directly on the exam (space will be provided)

Question 1: Short Answers (10 points out of a total of 100 points)

- (1a) [2 points] Given this set of data samples drawn from some Bayesian network over 4 binary variables Z, X, Y, W , what is the probability that W is true (t) given X is true (t) and Y is false (f) if using maximum likelihood learning of a binomial distribution.

Z	X	Y	W
t	t	f	f
t	f	t	f
t	t	f	t
t	t	t	t
t	t	f	f
t	f	f	t
t	f	t	t

SOLUTION:

The number of examples with $W=t$ and $X=t$ and $Y=f$ (1) divided by the number of examples where $X=t$ and $Y=f$ (3) : $2/3$

- (1b) [2 points] A heuristic for a search problem is an estimate of the true cost to the goal. Is the following statement true or false: *an admissible heuristic is always greater than the true cost.*

SOLUTION:

false

- (1c) [2 points] True or false: it is OK to ignore data with missing features/attributes when learning, but only if the data is *missing at random*

SOLUTION:

true

- (1d) [2 points] Complete the following sentence: A* search uses a priority queue of nodes ranked by _____, and so is a mixture of _____ and best-first search.

SOLUTION:

cost to the node + heuristic from node to goal, lowest-cost-first

- (1e) [2 points] True or False: Overfitting in machine learning refers to when your model has too much/little bias.

SOLUTION:

False/True

Question 2: Maximum Likelihood Learning (15 points out of a total of 100 points)

A *mixture of Gaussians* model is a Bayesian network $C \rightarrow X$ where C is a binary variable with values $\{1, 2\}$ and a binomial distribution $P(C) = \{\theta_c, 1 - \theta_c\}$ and X is a continuous variable with a Gaussian probability distribution:

$$P(x|c) = \frac{1}{\sqrt{2\pi}\sigma_c} e^{\frac{-(x-\mu_c)^2}{2\sigma_c^2}}$$

the parameters of this model are thus $\Theta = \{\theta_c, \mu_1, \sigma_1, \mu_2, \sigma_2\}$ where μ_i is the mean of the Gaussian (normal) distribution for category $c = i$ and σ_i is the variance of the Gaussian distribution for category $c = i$. You are given a dataset $\mathbf{d} = \{c_1, x_1, c_2, x_2, \dots, c_M, x_M\}$ giving complete data - one value for X and one value for C for each datapoint.

In this question, you will derive equations for the maximum likelihood estimates of the five parameters. Recall that this involves solving the following equation

$$\theta_{ML} = \arg \max_{\theta} P(\mathbf{d}|\Theta)$$

where $\theta \in \Theta$ is any parameter and θ_{ML} is the maximum likelihood estimate of that parameter. One does this by first writing down $P(\mathbf{d}|\Theta)$ using the binomial distribution $\theta_c^{c_j}(1 - \theta_c)^{1-c_j}$ and normal distribution $P(x_j|c_j)$ given above for data point j .

You may need the following derivatives

$$\begin{aligned}\frac{\partial}{\partial x} e^{f(x)} &= e^{f(x)} \frac{\partial}{\partial x} f(x) \\ \frac{\partial}{\partial x} \left(\frac{1}{f(x)^2} \right) &= -\frac{2}{f(x)^3} \frac{\partial}{\partial x} f(x) \\ \frac{\partial}{\partial x} \log(f(x)) &= \frac{1}{f(x)} \frac{\partial}{\partial x} f(x)\end{aligned}$$

(1a) [5 points] Derive an equation for the maximum likelihood estimate of θ_C .

SOLUTION:

take the derivative of $\log \prod_{j=1}^M \theta_c^{(c_j-1)} (1 - \theta_c)^{(2-c_j)} \frac{1}{\sqrt{2\pi}\sigma_{c_j}} e^{\frac{-(x_j-\mu_{c_j})^2}{2\sigma_{c_j}^2}}$ with respect to θ_c - this gives $\theta_c = \frac{1}{M} \sum_j (c_j - 1)$

(1b) [5 points] Derive an equation for the maximum likelihood estimate of μ_i .

SOLUTION:

take the derivative wrt μ_1

$$-\sum_{j|c_j=1} \frac{\partial}{\partial \mu_1} \frac{(x_j - \mu_1)^2}{2\sigma_1^2} = \sum_{j|c_j=1} \frac{(x_j - \mu_1)}{\sigma_1^2} = 0$$

which gives $\mu_1 = \frac{1}{M_1} \sum_{j|c_j=1} x_j$ where M_1 is the number of datapoints with $c_j = 1$.

(1c) [5 points] Derive an equation for the maximum likelihood estimate of σ_i .

SOLUTION:

take the derivative wrt σ_1

$$-\sum_{j|c_j=1} \frac{\partial}{\partial \sigma_1} \left[-\log \sqrt{2\pi} \sigma_1 + \frac{(x_j - \mu_1)^2}{2\sigma_1^2} \right] = \sum_{j|c_j=1} \frac{1}{\sigma_1} - \frac{(x - \mu_1)^2}{\sigma_1^3} = 0$$

which gives $\sigma_1^2 = \frac{1}{M_1} \sum_{j|c_j=1} (x_j - \mu_1)^2$ where M_1 is the number of datapoints with $c_j = 1$.

Question 3: Decision Trees (20 points out of a total of 100 points)

You are trying to predict who will win the Stupley Can™, annual prize handed out to the winner of the National Chokey League™ (Chokey is a team sport that is popular in Olertawo). There are 12 teams in the league and you have a database of historical information about previous winners and losers. Your database is a table where each row indicates a team's performance. The statistics you are given include the team's win percentage (binary variable $\in \{hi, lo\}$), whether they got more than 100 minutes of penalties/season on average $\in \{true, false\}$ and if the average number of goals they scored was greater than the team average for the league $\in \{true, false\}$. Finally, the last column gives the number of times this team has won the Stupley Can ($\in \{1, 2, 3\}$). Here is the dataset:

win %	>100mins penalties	>avg goals	# Stupley Can wins
lo	false	yes	3
hi	true	no	2
hi	true	yes	3
lo	false	no	1
lo	true	no	1
hi	true	no	1
hi	false	yes	3
lo	false	yes	3
hi	true	no	2
lo	true	no	2
lo	true	yes	3
lo	false	no	1

Recall that, given a set E of N training examples, if the number of examples with output (target) feature $Y = y_i$ is N_i , then

$$P(Y = y_i) = P(y_i) = \frac{N_i}{N}$$

and the total information content for the set E is:

$$I(E) = - \sum_{y_i \in Y} P(y_i) \log_2 P(y_i)$$

Using information gain (weighted by fractions of documents) as your feature selection mechanism, construct a decision tree that predicts how many times a team will win the Stupley Can based on the features of **win%**, **penalties** and **goals**. Continue splitting until no more attributes are available, or until all data agrees on the target attribute (**# of Stupley Can wins**). Break ties in feature selection arbitrarily. Use the majority class at each leaf as your point estimate (prediction). In case of a tie, choose the lower wins number. Show your work and draw a graph of your final decision tree.

SOLUTION:

(i) Round 1: Initial information is $-(5/12) \log(5/12) - (3/12) \log(3/12) - (4/12) \log(4/12) = 1.555$

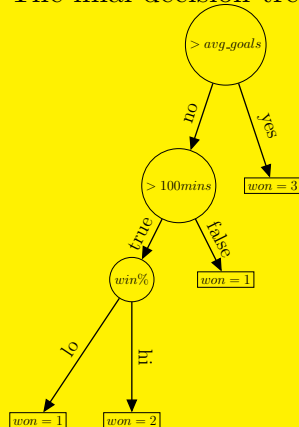
- win% has 7/5 split lo/hi. For win%=lo, 3,1,3 split so $I(3, 1, 3) = 1.45$, for win%=hi, 2/2/1 split, so $I(2, 2, 1) = 1.52$, so the $I(split) = (7/12) * 1.45 + (5/12) * 1.52 = 1.47$
- >100mins has 5/7 split false/true with information in the splits of $I(3, 0, 2) = 0.97$ and $I(2, 3, 2) = 1.56$ to total is 1.312
- >avg_goals has a 5/7 split y/n with $I(5, 0, 0) = 0$ and $I(0, 3, 4) = 0.985$ for total of 0.571 *the best split

(ii) Round 2: for >avg_goals=yes, performance is always 3, so this is a leaf. For >avg_goals=no

- win% has a 4/3 split with $I(0, 1, 3) = 0.81$ and $I(0, 2, 1) = 0.92$ total is 0.857
- >100mins has a 5/2 split with $I(0, 3, 2) = 0.97$ and $I(0, 0, 2) = 0$ total is 0.694 *the best split

(iii) Round 3: for >100mins=no, performance is always 1 so this is a leaf, For >100mins=yes the last split on win% gives a perfect prediction for *hi* of 2 and a 50/50 split for lo so either works but we choose the lower one in case of a tie so its a 1

The final decision tree is therefore:



Question 4: Neural Networks (30 points out of a total of 100 points)

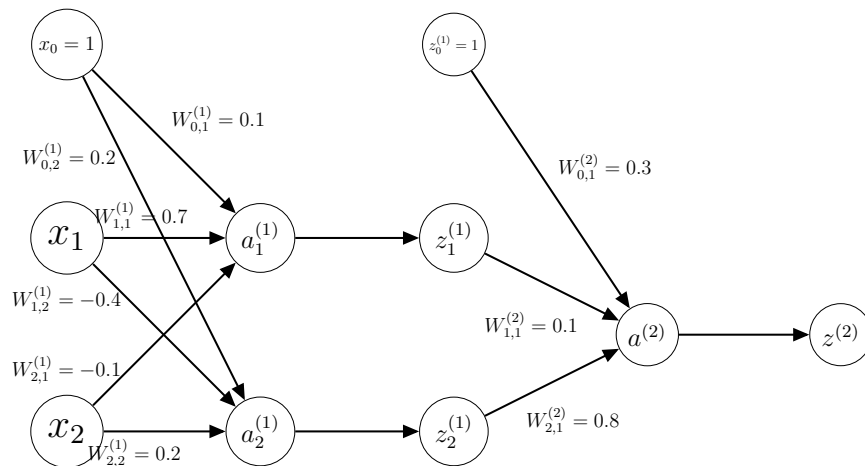
Carolyn is a machine learning engineer working with a soccer team. Carolyn would like to predict whether the team will win their next game. She therefore treats this as a binary classification problem. After querying the team's records, Carolyn assembles a dataset consisting of previous games. Each example consists of features regarding the game, and the target is whether or not her team won the game. The features and target are:

Features:

- x_1 : The game number (out of the total games for a season)
- x_2 : The number of injured players

Target: Whether the team won or lost the game.

Carolyn trains the following neural network, using a sigmoid $f(x) = \frac{1}{1+e^{-x}}$ as the activation function for both hidden layers. The network is trained to minimize the squared error function (for each datapoint): $E(\hat{y}, y) = (y - \hat{y})^2$



Recall that, starting with l , the error is computed for the output layer (l) as:

$$\vec{\delta}^{(l)} = \frac{\partial Error}{\partial \vec{z}^{(l)}} \times f'(\vec{a}^{(l)}) \quad (1)$$

and the error for the other layers starting from $l - 1$ (next to last layer):

$$\vec{\delta}^{(l)} = \left(\vec{\delta}^{(l+1)} \cdot \mathbf{W}^{(l+1)T} \right) \times f'(\vec{a}^{(l)}) \quad (2)$$

then the weight updates for the input layer are:

$$\frac{\partial Error}{\partial \mathbf{W}^{(l)}} = \vec{x}^T \vec{\delta}^{(l)} \quad (3)$$

and other layers:

$$\frac{\partial Error}{\partial \mathbf{W}^{(l)}} = \mathbf{z}^{(l-1)T} \delta^{(l)} \quad (4)$$

and finally update the weights are updated using:

$$\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta \frac{\partial Error}{\partial \mathbf{W}^{(l)}}$$

- 3(a) [8 points] Execute the forward pass of this neural network for the example $x = \langle 1, 2 \rangle$. Use the weights that are labelled on the diagram above. The final output should be the network's prediction, $\hat{y} = z^{(2)}$.

SOLUTION:

$$\begin{aligned} a1 &= [1 \quad 1 \quad 2] \begin{bmatrix} 0.1 & 0.2 \\ 0.7 & -0.4 \\ -0.1 & 0.2 \end{bmatrix} = [0.6 \quad 0.2] \\ z1 &= f(a1) = [0.645 \quad 0.549] \\ a2 &= [1 \quad 0.645 \quad 0.549] \begin{bmatrix} 0.3 \\ 0.1 \\ 0.8 \end{bmatrix} = [0.804] \\ z2 &= \frac{1}{1 + e^{-0.804}} = 0.691 \end{aligned}$$

- 3(b) [2 points] Suppose that the target for example $x = \langle 1, 2 \rangle$ is $y = 1$ (game was won). What is the value of the error function, given the network's prediction from part (a)?

SOLUTION:

$$(1 - 0.691)^2 = 0.0955$$

- 3(c) [17 points] Calculate the partial derivative of the error with respect to $W_{1,1}^{(2)}$ (i.e. $\frac{\partial E}{\partial W_{1,1}^{(2)}}$). Be sure to show all of your steps.

SOLUTION:

first calculate the derivative of the error

$$-2 * (1 - 0.691) = -0.618$$

$$\text{then calculate } \delta^{(2)} = -0.618 * f(0.804) * (1 - f(0.804)) = -0.132$$

now calculate $\frac{\partial E}{\partial W^{(2)}} = z^{(1)} * \delta^{(2)} = \begin{bmatrix} 1 \\ 0.645 \\ 0.549 \end{bmatrix} [-0.132] = \begin{bmatrix} -0.132 \\ -0.085 \\ -0.073 \end{bmatrix}$
so the answer is -0.085

- 3(d) [**3 points**] Using the partial derivative you calculated in part (a), update the value of the weight $W_{1,1}^{(2)}$. Use a learning rate of $\eta = 0.1$.

SOLUTION:

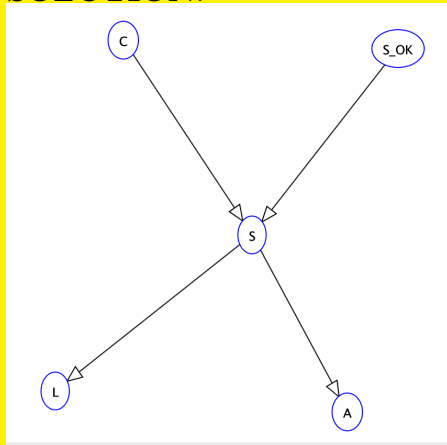
$$0.1 - 0.1 * -0.085 = 0.1085$$

Question 5: Bayesian Networks (25 points out of a total of 100 points)

A binary sensor (S) is used to detect overheating of the core in a nuclear reactor. The sensor may be working properly ($S_{ok}=true$) or malfunctioning ($S_{ok}=false$). If the reactor core overheats ($C = true$), and the sensor is working properly, the sensor will read $S=true$ with certainty (probability 1.0). If the core doesn't overheat, and the sensor is working properly, the sensor reads $S=true$ 3% of the time (false positives). If the sensor is malfunctioning then the sensor will read $S=true$ with probability 0.5 regardless of the state of the core. A positive reading on the sensor ($S=true$) causes an alarm to go off ($A=true$) 95% of the time, and causes all the doors in the reactor to lock ($L=true$) 99% of the time. If $S=false$, the alarm never goes off, but the doors lock with probability 0.02 . The sensor malfunctions with probability 0.01 , and the reactor core has a 10% chance of overheating.

- (3a) [5 points] Draw a Bayesian network representing the knowledge you have about the nuclear reactor above.

SOLUTION:



- (3b) [10 points] Write down a complete set of factors representing all the knowledge about the conditional probabilities in the BN. You should have 5 factors.

SOLUTION:

$$f_0(C) = P(C) = [0.1, 0.9]$$
$$f_1(S_{ok}) = P(S_{ok}) = [0.99, 0.01]$$

	C	S_{ok}	$P(S C, S_{ok})$
	T	T	$[1.0, 0.0]$
$f_2(S, C, S_{ok}) = P(S C, S_{ok}) =$	T	F	$[0.5, 0.5]$
	F	T	$[0.03, 0.97]$
	F	F	$[0.5, 0.5]$

	S	$P(A S)$
$f_3(A, S) = P(A S) =$	T	$[0.95, 0.05]$
	F	$[0.0, 1.0]$

	S	$P(L S)$
$f_4(L, S) = P(L S) =$	T	$[0.99, 0.01]$
	F	$[0.02, 0.98]$

Answer the following queries about the nuclear reactor using your Bayesian network and Variable Elimination. In the following questions, you can leave your answer as a product and/or sum of the real numbers found in your BN (i.e. you don't need to explicitly do the computations on your calculator, but the final answer should be in a form that could be put into a calculator, without any variables and Σ symbols).

- (3c) [2 points] What is the probability of the reactor core overheating (given no evidence)?

SOLUTION:

Everything is irrelevant, so $P(C = true) = f_0(C = true) = 0.1$

- (3d) [8 points] What is the probability that the reactor core is overheating given that the alarm is going off?

SOLUTION:

$$\begin{aligned}
& P(C = \text{true} | A = \text{true}) \\
&= \sum_{S, S_{ok}} P(C = \text{true}, S, S_{ok} | A = \text{true}) \\
&\propto \sum_{S, S_{ok}} f_0(C = \text{true}) f_1(S_{ok}) f_2(S, C = \text{true}, S_{ok}) f_3(A = \text{true}, S) \\
&= f_0(C = \text{true}) \sum_{S_{ok}} f_1(S_{ok}) \sum_S f_2(S, C = \text{true}, S_{ok}) f_3(A = \text{true}, S) \\
&= 0.1 \times [0.99 \times [1.0 \times 0.95] + 0.01 \times [0.5 \times 0.95 + 0.5 \times 0.0]] \\
&= 0.1 \times [0.99 \times 0.95 + 0.01 \times 0.475] = 0.094525 \\
& P(C = \text{false} | A = \text{true}) \\
&= f_0(C = \text{false}) \sum_{S_{ok}} f_1(S_{ok}) \sum_S f_2(S, C = \text{false}, S_{ok}) f_3(A = \text{true}, S) \\
&= 0.9 \times [0.99 \times [0.03 \times 0.95 + 0.97 \times 0.0] + 0.01 \times [0.5 \times 0.95 + 0.5 \times 0.0]] \\
&= 0.9 \times [0.99 \times 0.0285 + 0.01 \times 0.475] = 0.0296685
\end{aligned}$$

So normalised is

$$\begin{aligned}
& P(C = \text{true} | A = \text{true}) \\
&= \frac{0.1 \times [0.99 \times 0.95 + 0.01 \times 0.45]}{0.1 \times [0.99 \times 0.95 + 0.01 \times 0.475] + 0.9 \times [0.99 \times 0.0285 + 0.01 \times 0.475]} \\
&= 0.7611
\end{aligned}$$